

Lab - Working with Database in Python Programming

1. Create the following tables:

Table name: CLIENT_MASTER

Column Name	Data Type	Size	Attributes
CLIENTNO	Varchar	6	Primary Key / First letter must start with 'C'
NAME	Varchar	20	Not Null
CITY	Varchar	15	
PINCODE	Number	8	
STATE	Varchar	15	
BALDUE	Number	10,2	

```
mysql> use testdb;
Database changed
mysql> DESCRIBE CLIENT_MASTER;
+-----+-----+-----+-----+-----+-----+
| Field      | Type           | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CLIENTNO   | varchar(6)     | NO   | PRI | NULL    |       |
| NAME       | varchar(20)    | NO   |     | NULL    |       |
| CITY       | varchar(15)    | YES  |     | NULL    |       |
| PINCODE    | int            | YES  |     | NULL    |       |
| STATE      | varchar(15)    | YES  |     | NULL    |       |
| BALDUE     | decimal(10,2) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.01 sec)
```

Table_Name: PRODUCT_MASTER

Column Name	Data Type	Size	Attributes
PRODUCTNO	Varchar	6	Primary Key / First letter must start with 'P'
DESCRIPTION	Varchar	15	Not Null
PROFITPERCENT	Number	4,2	Not Null
UNITMEASURE	Varchar	10	Not Null
QTYONHAND	Number	8	Not Null
REORDERLVL	Number	8	Not Null
SELLPRICE	Number	8,2	Not Null, cannot be 0
COSTPRICE	Number	8,2	Not Null, cannot be 0

```
mysql> CREATE TABLE PRODUCT_MASTER (  
-> PRODUCTNO VARCHAR(6) PRIMARY KEY CHECK (PRODUCTNO LIKE 'P%'),  
-> DESCRIPTION VARCHAR(15) NOT NULL,  
-> PROFITPERCENT DECIMAL(4, 2) NOT NULL,  
-> UNITMEASURE VARCHAR(10) NOT NULL,  
-> QTYONHAND INT NOT NULL,  
-> REORDERLVL INT NOT NULL,  
-> SELLPRICE DECIMAL(8, 2) NOT NULL CHECK (SELLPRICE > 0),  
-> COSTPRICE DECIMAL(8, 2) NOT NULL CHECK (COSTPRICE > 0)  
-> );
```

```
mysql> DESCRIBE PRODUCT_MASTER;  
+-----+-----+-----+-----+-----+-----+  
| Field          | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| PRODUCTNO      | varchar(6)    | NO   | PRI | NULL    |       |  
| DESCRIPTION     | varchar(15)   | NO   |     | NULL    |       |  
| PROFITPERCENT  | decimal(4,2)  | NO   |     | NULL    |       |  
| UNITMEASURE    | varchar(10)   | NO   |     | NULL    |       |  
| QTYONHAND      | int           | NO   |     | NULL    |       |  
| REORDERLVL     | int           | NO   |     | NULL    |       |  
| SELLPRICE      | decimal(8,2)  | NO   |     | NULL    |       |  
| COSTPRICE      | decimal(8,2)  | NO   |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
8 rows in set (0.00 sec)
```

Table_Name: SALESMAN_MASTER

Column Name	Data Type	Size	Attributes
SALESMANNO	Varchar	6	Primary Key / First letter must start with 'S'
SALESMANNAME	Varchar	20	Not Null
ADDRESS1	Varchar	30	Not Null
ADDRESS2	Varchar	30	
CITY	Varchar	20	
PINCODE	Number	8	
STATE	Varchar	20	
SALAMT	Number	8,2	Not Null, cannot be 0
TGTTOGET	Number	6,2	Not Null, cannot be 0
YTDSALES	Number	6,2	Not Null
REMARKS	Varchar	60	

```
mysql> CREATE TABLE SALESMAN_MASTER (
-> SALESMANNO VARCHAR(6) PRIMARY KEY CHECK (SALESMANNO LIKE 'S%'),
-> SALESMANNAME VARCHAR(20) NOT NULL,
-> ADDRESS1 VARCHAR(30) NOT NULL,
-> ADDRESS2 VARCHAR(30),
-> CITY VARCHAR(20),
-> PINCODE INT,
-> STATE VARCHAR(20),
-> SALAMT DECIMAL(8, 2) NOT NULL CHECK (SALAMT > 0),
-> TGTTOGET DECIMAL(6, 2) NOT NULL CHECK (TGTTOGET > 0),
-> YTDSALES DECIMAL(6, 2) NOT NULL,
-> REMARKS VARCHAR(60)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> DESCRIBE SALESMAN_MASTER;
```

Field	Type	Null	Key	Default	Extra
SALESMANNO	varchar(6)	NO	PRI	NULL	
SALESMANNAME	varchar(20)	NO		NULL	
ADDRESS1	varchar(30)	NO		NULL	
ADDRESS2	varchar(30)	YES		NULL	
CITY	varchar(20)	YES		NULL	
PINCODE	int	YES		NULL	
STATE	varchar(20)	YES		NULL	
SALAMT	decimal(8,2)	NO		NULL	
TGTTOGET	decimal(6,2)	NO		NULL	
YTDSALES	decimal(6,2)	NO		NULL	
REMARKS	varchar(60)	YES		NULL	

Table_Name: SALES_ORDER

Column Name	Data Type	Size	Attributes
ORDERNO	Varchar	6	Primary Key / First letter must start with 'O'
CLIENTNO	Varchar	6	Foreign Key references ClientNo of Client_Master table
ORDERDATE	Date		Not Null
DELYADDR	Varchar	25	
SALESMANNO	Varchar	6	Foreign Key references SalesmanNo of Salesman_Master table
DELYTYPE	Char	1	
BILLYN	Char	1	
DELYDATE	Date		
ORDERSTATUS	Varchar	10	

```
mysql> CREATE TABLE SALES_ORDER (
-> ORDERNO VARCHAR(6) PRIMARY KEY CHECK (ORDERNO LIKE 'O%'),
-> CLIENTNO VARCHAR(6) NOT NULL,
-> ORDERDATE DATE NOT NULL,
-> DELYADDR VARCHAR(25),
-> SALESMANNO VARCHAR(6),
-> DELYTYPE CHAR(1),
-> BILLYN CHAR(1),
-> DELYDATE DATE,
-> ORDERSTATUS VARCHAR(10),
-> FOREIGN KEY (CLIENTNO) REFERENCES CLIENT_MASTER(CLIENTNO),
-> FOREIGN KEY (SALESMANNO) REFERENCES SALESMAN_MASTER(SALESMANNO)
-> );
Query OK, 0 rows affected (0.07 sec)

mysql> DESCRIBE SALES_ORDER;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ORDERNO    | varchar(6) | NO   | PRI | NULL    |       |
| CLIENTNO    | varchar(6) | NO   | MUL | NULL    |       |
| ORDERDATE  | date       | NO   |     | NULL    |       |
| DELYADDR    | varchar(25) | YES  |     | NULL    |       |
| SALESMANNO | varchar(6) | YES  | MUL | NULL    |       |
| DELYTYPE    | char(1)    | YES  |     | NULL    |       |
| BILLYN     | char(1)    | YES  |     | NULL    |       |
| DELYDATE    | date       | YES  |     | NULL    |       |
| ORDERSTATUS | varchar(10) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

Table_Name: SALES_ORDER_DETAILS

Column Name	Data Type	Size	Attributes
ORDERNO	Varchar	6	Foreign Key references OrderNo of Sales_Order table
PRODUCTNO	Varchar	6	Foreign Key references ProductNo of Product_Master table
QTYORDERED	Number	8	
QTYDISP	Number	8	
PRODUCTRATE	Number	10,2	

```
mysql> CREATE TABLE SALES_ORDER_DETAILS (  
-> ORDERNO VARCHAR(6) NOT NULL,  
-> PRODUCTNO VARCHAR(6) NOT NULL,  
-> QTYORDERED INT(8),  
-> QTYDISP INT(8),  
-> PRODUCTRATE DECIMAL(10, 2),  
-> FOREIGN KEY (ORDERNO) REFERENCES SALES_ORDER(ORDERNO),  
-> FOREIGN KEY (PRODUCTNO) REFERENCES PRODUCT_MASTER(PRODUCTNO)  
-> );  
Query OK, 0 rows affected, 2 warnings (0.07 sec)  
  
mysql> DESCRIBE SALES_ORDER_DETAILS;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| ORDERNO    | varchar(6)    | NO   | MUL | NULL    |      |  
| PRODUCTNO  | varchar(6)    | NO   | MUL | NULL    |      |  
| QTYORDERED | int           | YES  |     | NULL    |      |  
| QTYDISP    | int           | YES  |     | NULL    |      |  
| PRODUCTRATE | decimal(10,2) | YES  |     | NULL    |      |  
+-----+-----+-----+-----+-----+-----+  
5 rows in set (0.00 sec)
```

2. Insert the following data into their respective tables:

Data for CLIENT_MASTER table:

CLIENT	NAME	CITY	PINCODE	STATE	BALDUE
C00001	Ivan Bayross	Mumbai	400054	Maharashtra	15000
C00003	Chhaya Bankar	Mumbai	400057	Maharashtra	5000
C00004	Ashwini Joshi	Bangalore	560001	Karnataka	0
C00005	Hansel Colaco	Mumbai	400060	Maharashtra	2000
C00006	Deepak Sharma	Mangalore	560050	Karnataka	0

```
mysql> INSERT INTO CLIENT_MASTER (CLIENTNO, NAME, CITY, PINCODE, STATE, BALDUE) VALUES
-> ('C00001', 'Ivan Bayross', 'Mumbai', '400054', 'Maharashtra', 15000.00),
-> ('C00003', 'Chhaya Bankar', 'Mumbai', '400057', 'Maharashtra', 5000.00),
-> ('C00004', 'Ashwini Joshi', 'Bangalore', '560001', 'Karnataka', 0.00),
-> ('C00005', 'Hansel Colaco', 'Mumbai', '400060', 'Maharashtra', 2000.00),
-> ('C00007', 'Deepak Sharma', 'Mangalore', '560050', 'Karnataka', 0.00);
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM CLIENT_MASTER;
```

CLIENTNO	NAME	CITY	PINCODE	STATE	BALDUE
C00001	Ivan Bayross	Mumbai	400054	Maharashtra	15000.00
C00003	Chhaya Bankar	Mumbai	400057	Maharashtra	5000.00
C00004	Ashwini Joshi	Bangalore	560001	Karnataka	0.00
C00005	Hansel Colaco	Mumbai	400060	Maharashtra	2000.00
C00007	Deepak Sharma	Mangalore	560050	Karnataka	0.00

```
5 rows in set (0.00 sec)
```

Data for PRODUCT_MASTER table

PRODUC	DESCRIPTION	PROFITPERCENT	UNITMEASUR	QTYONHAND	REORDERLVL	SELLPRICE	COSTPRICE
P00001	T-Shirts	5	Piece	200	50	350	250
P03453	Shirts	6	Piece	150	50	500	350
P06734	Cotton Jeans	5	Piece	100	20	600	450
P07865	Jeans	5	Piece	100	20	750	500
P07868	Trousers	2	Piece	150	50	850	550
P07885	Pull Overs	2.5	Piece	80	30	700	450
P07965	Denim Shirts	4	Piece	100	40	350	250
P07975	Lycra Tops	5	Piece	70	30	300	175
P08865	Skirts	5	Piece	75	30	450	300

```
mysql> INSERT INTO PRODUCT_MASTER (PRODUCTNO, DESCRIPTION, PROFITPERCENT, UNITMEASURE, QTYONHAND, REORDERLVL, SELLPRICE, COSTPRICE) V
VALUES
-> ('P00001', 'T-Shirts', 5, 'Piece', 200, 50, 350, 250),
-> ('P03453', 'Shirts', 6, 'Piece', 150, 50, 500, 350),
-> ('P06734', 'Cotton Jeans', 5, 'Piece', 100, 20, 600, 450),
-> ('P07865', 'Jeans', 5, 'Piece', 100, 20, 750, 500),
-> ('P07868', 'Trousers', 2, 'Piece', 150, 50, 850, 550),
-> ('P07885', 'Pull Overs', 2.5, 'Piece', 80, 30, 700, 450),
-> ('P07965', 'Denim Shirts', 4, 'Piece', 100, 40, 350, 250),
-> ('P07975', 'Lycra Tops', 5, 'Piece', 70, 30, 300, 175),
-> ('P08865', 'Skirts', 5, 'Piece', 75, 30, 450, 300);
Query OK, 9 rows affected (0.02 sec)
Records: 9 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM PRODUCT_MASTER;
```

PRODUCTNO	DESCRIPTION	PROFITPERCENT	UNITMEASURE	QTYONHAND	REORDERLVL	SELLPRICE	COSTPRICE
P00001	T-Shirts	5.00	Piece	200	50	350.00	250.00
P03453	Shirts	6.00	Piece	150	50	500.00	350.00
P06734	Cotton Jeans	5.00	Piece	100	20	600.00	450.00
P07865	Jeans	5.00	Piece	100	20	750.00	500.00
P07868	Trousers	2.00	Piece	150	50	850.00	550.00
P07885	Pull Overs	2.50	Piece	80	30	700.00	450.00
P07965	Denim Shirts	4.00	Piece	100	40	350.00	250.00
P07975	Lycra Tops	5.00	Piece	70	30	300.00	175.00
P08865	Skirts	5.00	Piece	75	30	450.00	300.00

```
9 rows in set (0.00 sec)
```

Data for SALESMAN_MASTER table

SALESM	SALESMANNAME	ADDRESS1	ADDRESS2	CITY	PINCODE	STATE	SALAMT	TGTTGET	YTDSALES	REMARKS
S00001	Aman	A/14	Worli	Mumbai	400002	Maharashtra	3000	100	50	Good
S00002	Omkar	65	Nariman	Mumbai	400001	Maharashtra	3000	200	100	Good
S00003	Raj	P-7	Bandra	Mumbai	400032	Maharashtra	3000	200	100	Good
S00004	Ashish	A/5	Juhu	Bombay	400044	Maharashtra	3500	200	150	Good

```
mysql> INSERT INTO SALESMAN_MASTER (SALESMANNO, SALESMANNAME, ADDRESS1, ADDRESS2, CITY, PINCODE, STATE, SALAMT, TGTTGET, YTDSALES, R
EMARKS)
-> VALUES
-> ('S00001', 'Aman', 'A/14', 'Worli', 'Mumbai', '400002', 'Maharashtra', 3000, 100, 50, 'Good'),
-> ('S00002', 'Omkar', '65', 'Nariman', 'Mumbai', '400001', 'Maharashtra', 3000, 200, 100, 'Good'),
-> ('S00003', 'Raj', 'P-7', 'Bandra', 'Mumbai', '400032', 'Maharashtra', 3000, 200, 100, 'Good'),
-> ('S00004', 'Ashish', 'A/5', 'Juhu', 'Bombay', '400044', 'Maharashtra', 3500, 200, 150, 'Good');
Query OK, 4 rows affected (0.02 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM SALESMAN_MASTER;
```

SALESMANNO	SALESMANNAME	ADDRESS1	ADDRESS2	CITY	PINCODE	STATE	SALAMT	TGTTGET	YTDSALES	REMARKS
S00001	Aman	A/14	Worli	Mumbai	400002	Maharashtra	3000.00	100.00	50.00	Good
S00002	Omkar	65	Nariman	Mumbai	400001	Maharashtra	3000.00	200.00	100.00	Good
S00003	Raj	P-7	Bandra	Mumbai	400032	Maharashtra	3000.00	200.00	100.00	Good
S00004	Ashish	A/5	Juhu	Bombay	400044	Maharashtra	3500.00	200.00	150.00	Good

```
4 rows in set (0.00 sec)
```

Data for Sales_Order Table

ORDERN	CLIENT	ORDERDATE	DELYADDR	SALESM	D	B	DELYDATE	ORDERSTATU
019003	C00001	03-APR-02	Delhi	S00001	F	Y	07-APR-02	Fulfilled
046866	C00004	20-MAY-02	Delhi	S00002	P	N	22-MAY-02	Cancelled
019008	C00005	24-MAY-02	Delhi	S00004	F	N	26-JUL-96	In Process
019001	C00001	12-JUN-02	Delhi	S00001	F	N	20-JUL-02	In Process
019002	C00002	25-JUN-02	Delhi	S00002	P	N	27-JUL-02	Cancelled

```
mysql> INSERT INTO Sales_Order (ORDERNO, CLIENTNO, ORDERDATE, DELYADDR, SALESMANNO, DELYTYPE, BILLYN, DELYDATE, ORDERSTATUS)
-> VALUES
-> ('019003', 'C00001', '2002-04-03', 'Delhi', 'S00001', 'F', 'Y', '2002-04-07', 'Fulfilled'),
-> ('046866', 'C00004', '2002-05-20', 'Delhi', 'S00002', 'P', 'Y', '2002-05-22', 'Cancelled'),
-> ('019008', 'C00005', '2002-05-24', 'Delhi', 'S00004', 'F', 'Y', '1996-07-26', 'In Process'),
-> ('019001', 'C00001', '2002-06-12', 'Delhi', 'S00001', 'F', 'Y', '2002-07-20', 'In Process'),
-> ('019002', 'C00002', '2002-06-25', 'Delhi', 'S00002', 'P', 'Y', '2002-07-27', 'Cancelled');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM SALES_ORDER;
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| ORDERNO | CLIENTNO | ORDERDATE | DELYADDR | SALESMANNO | DELYTYPE | BILLYN | DELYDATE | ORDERSTATUS |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 019001 | C00001 | 2002-06-12 | Delhi | S00001 | F | Y | 2002-07-20 | In Process |
| 019002 | C00002 | 2002-06-25 | Delhi | S00002 | P | Y | 2002-07-27 | Cancelled |
| 019003 | C00001 | 2002-04-03 | Delhi | S00001 | F | Y | 2002-04-07 | Fulfilled |
| 019008 | C00005 | 2002-05-24 | Delhi | S00004 | F | Y | 1996-07-26 | In Process |
| 046866 | C00004 | 2002-05-20 | Delhi | S00002 | P | Y | 2002-05-22 | Cancelled |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Data for Sales_Order_Details Table

OrderNo	ProductNo	Qtyordered	QtyDisp	ProductRate
019001	P00001	4	4	525
019001	P07965	2	1	8400
019001	P07885	2	1	5250
019002	P00001	10	0	525
046865	P07868	3	3	3150
046865	P07885	3	1	5250
046865	P00001	10	10	525
046865	P03453	4	4	1050
019003	P03453	2	2	1050
019003	P06734	1	1	12000
046866	P07965	1	0	8400
046866	P07975	1	0	1050
019008	P00001	10	5	525
019008	P07975	5	3	1050

```
mysql> INSERT INTO Sales_Order_Details (OrderNo, ProductNo, QtyOrdered, QtyDisp, ProductRate)
-> VALUES
-> ('019001', 'P00001', 4, 4, 525),
-> ('019001', 'P07965', 2, 1, 8400),
-> ('019001', 'P07885', 2, 1, 5250),
-> ('019002', 'P00001', 10, 0, 525),
-> ('046865', 'P07868', 3, 3, 3150),
-> ('046865', 'P07885', 3, 1, 5250),
-> ('046865', 'P00001', 10, 10, 525),
-> ('046865', 'P03453', 4, 4, 1050),
-> ('019003', 'P03453', 2, 2, 1050),
-> ('019003', 'P06734', 1, 1, 12000),
-> ('046866', 'P07965', 1, 0, 8400),
-> ('046866', 'P07975', 1, 0, 1050),
-> ('019008', 'P00001', 10, 5, 525),
-> ('019008', 'P07975', 5, 3, 1050);
Query OK, 14 rows affected (0.01 sec)
Records: 14 Duplicates: 0 Warnings: 0
```



```
mysql> SELECT * FROM SALES_ORDER_DETAILS;
```

ORDERNO	PRODUCTNO	QTYORDERED	QTYDISP	PRODUCTRATE
019001	P00001	4	4	525.00
019001	P07965	2	1	8400.00
019001	P07885	2	1	5250.00
019002	P00001	10	0	525.00
046865	P07868	3	3	3150.00
046865	P07885	3	1	5250.00
046865	P00001	10	10	525.00
046865	P03453	4	4	1050.00
019003	P03453	2	2	1050.00
019003	P06734	1	1	12000.00
046866	P07965	1	0	8400.00
046866	P07975	1	0	1050.00
019008	P00001	10	5	525.00
019008	P07975	5	3	1050.00

14 rows in set (0.00 sec)

3. Generate the SQL statements to perform the following computations on table data in Python IDLE:

- a. List the names of all clients having 'a' as the second letter in their names.

```
mysql> SELECT name
-> FROM client_master
-> WHERE name LIKE '_a%';
```

name
Hansel Colaco

1 row in set (0.01 sec)

```
import mysql.connector

try:
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="cdac123",
        database="testdb",
    )
    cursor = conn.cursor()

    cursor.execute("SELECT name FROM client_master WHERE name LIKE '_a%';")
    results = cursor.fetchall()

    for row in results:
        print(row)
except mysql.connector.Error as err:
    print(f"Error: {err}")
finally:
    if 'cursor' in locals():
        cursor.close()
    if 'conn' in locals() and conn.is_connected():
        conn.close()

IDLE Shell 3.12.6
File Edit Shell Debug Options Window Help
Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/client_masterA.py
('Hansel Colaco',)
```

b. List the clients who stay in a city whose first letter is 'M'.

```
mysql> SELECT *
-> FROM client_master
-> WHERE city LIKE 'M%';
```

CLIENTNO	NAME	CITY	PINCODE	STATE	BALDUE
C00001	Ivan Bayross	Mumbai	400054	Maharashtra	15000.00
C00003	Chhaya Bankar	Mumbai	400057	Maharashtra	5000.00
C00005	Hansel Colaco	Mumbai	400060	Maharashtra	2000.00
C00007	Deepak Sharma	Mangalore	560050	Karnataka	0.00

```
4 rows in set (0.00 sec)
```

```
import mysql.connector

try:
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="cdac123",
        database="testdb"
    )
    cursor = conn.cursor()

    cursor.execute("SELECT * FROM client_master WHERE city LIKE 'M%';")
    results = cursor.fetchall()

    for row in results:
        print(row)

except mysql.connector.Error as err:
    print(f"Error: {err}")
finally:
    if 'cursor' in locals():
        cursor.close()
    if 'conn' in locals() and conn.is_connected():
        conn.close()
```

```
IDLE Shell 3.12.6
File Edit Shell Debug Options Window Help
Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/client_masterA.py
('C00001', 'Ivan Bayross', 'Mumbai', 400054, 'Maharashtra', Decimal('15000.00'))
('C00003', 'Chhaya Bankar', 'Mumbai', 400057, 'Maharashtra', Decimal('5000.00'))
('C00005', 'Hansel Colaco', 'Mumbai', 400060, 'Maharashtra', Decimal('2000.00'))
('C00007', 'Deepak Sharma', 'Mangalore', 560050, 'Karnataka', Decimal('0.00'))
>>>
```

c. List all clients who stay in 'Bangalore' or 'Mangalore'.

```
mysql> SELECT *
-> FROM client_master
-> WHERE city IN ('Bangalore', 'Mangalore');
```

CLIENTNO	NAME	CITY	PINCODE	STATE	BALDUE
C00004	Ashwini Joshi	Bangalore	560001	Karnataka	0.00
C00007	Deepak Sharma	Mangalore	560050	Karnataka	0.00

```
2 rows in set (0.00 sec)
```

```
File Edit Format Run Options Window Help
import mysql.connector

try:
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="cdac123",
        database="testdb"
    )
    cursor = conn.cursor()

    cursor.execute("SELECT * FROM client_master WHERE city IN ('Bangalore', 'Mangalore');")
    results = cursor.fetchall()

    for row in results:
        print(row)

except mysql.connector.Error as err:
    print(f"Error: {err}")
finally:
    if 'cursor' in locals():
        cursor.close()
    if 'conn' in locals() and conn.is_connected():
        conn.close()
```

IDLE Shell 3.12.6

File Edit Shell Debug Options Window Help

Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)]
Type "help", "copyright", "credits" or "license()" for more information.

```
>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/client_masterA.py
('C00004', 'Ashwini Joshi', 'Bangalore', 560001, 'Karnataka', Decimal('0.00'))
('C00007', 'Deepak Sharma', 'Mangalore', 560050, 'Karnataka', Decimal('0.00'))
...

```

d. List all clients whose BalDue is greater than value 10000.

```
mysql> SELECT *
-> FROM client_master
-> WHERE baldue > 1000;
```

CLIENTNO	NAME	CITY	PINCODE	STATE	BALDUE
C00001	Ivan Bayross	Mumbai	400054	Maharashtra	15000.00
C00003	Chhaya Bankar	Mumbai	400057	Maharashtra	5000.00
C00005	Hansel Colaco	Mumbai	400060	Maharashtra	2000.00

3 rows in set (0.00 sec)

```
import mysql.connector

try:
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="cdac123",
        database="testdb"
    )
    cursor = conn.cursor()

    cursor.execute("SELECT * FROM client_master WHERE baldue > 10000;")
    results = cursor.fetchall()

    for row in results:
        print(row)

except mysql.connector.Error as err:
    print(f"Error: {err}")
finally:
    if 'cursor' in locals():
        cursor.close()
    if 'conn' in locals() and conn.is_connected():
        conn.close()
```

IDLE Shell 3.12.6

File Edit Shell Debug Options Window Help

Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)]
Type "help", "copyright", "credits" or "license()" for more information.

```
>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/client_masterA.py
('C00001', 'Ivan Bayross', 'Mumbai', 400054, 'Maharashtra', Decimal('15000.00'))
('C00003', 'Chhaya Bankar', 'Mumbai', 400057, 'Maharashtra', Decimal('5000.00'))
('C00005', 'Hansel Colaco', 'Mumbai', 400060, 'Maharashtra', Decimal('2000.00'))
...

```

e. List all information from the Sales_order table for order placed in the month of June.

```
mysql> SELECT *
-> FROM Sales_order
-> WHERE MONTH(orderdate)=6;
```

ORDERNO	CLIENTNO	ORDERDATE	DELYADDR	SALESMANNO	DELYTYPE	BILLYN	DELYDATE	ORDERSTATUS
019001	C00001	2002-06-12	Delhi	S00001	F	Y	2002-07-20	In Process
019002	C00002	2002-06-25	Delhi	S00002	P	Y	2002-07-27	Cancelled

2 rows in set (0.01 sec)

```
import mysql.connector

try:
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="cdac123",
        database="testdb"
    )
    cursor = conn.cursor()

    cursor.execute("SELECT * FROM Sales_order WHERE MONTH(orderdate) = 6;")
    results = cursor.fetchall()

    for row in results:
        print(row)

except mysql.connector.Error as err:
    print(f"Error: {err}")
finally:
    if 'cursor' in locals():
        cursor.close()
    if 'conn' in locals() and conn.is_connected():
        conn.close()
```

IDLE Shell 3.12.6

Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

```
>>> ===== RESTART: C:/Users/HP/OneDrive/Desktop/client_masterA.py =====
('019001', 'C00001', datetime.date(2002, 6, 12), 'Delhi', 'S00001', 'F', 'Y', datetime.date(2002, 7, 20), 'In Process')
('019002', 'C00002', datetime.date(2002, 6, 25), 'Delhi', 'S00002', 'P', 'Y', datetime.date(2002, 7, 27), 'Cancelled')
>>>
```

f. List the Order No & day on which clients placed their order.

```
mysql> SELECT orderno, DATE(orderdate) AS order_day
-> FROM Sales_order;
```

orderno	order_day
019001	2002-06-12
019002	2002-06-25
019003	2002-04-03
019008	2002-05-24
046865	2002-05-20
046866	2002-05-20

6 rows in set (0.00 sec)

```

import mysql.connector

try:
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="cdac123",
        database="testdb"
    )
    cursor = conn.cursor()

    cursor.execute("SELECT orderno, DATE(orderdate) AS order_day FROM Sales_order;")
    results = cursor.fetchall()

    for row in results:
        print(row)

except mysql.connector.Error as err:
    print(f"Error: {err}")
finally:
    if 'cursor' in locals():
        cursor.close()
    if 'conn' in locals() and conn.is_connected():
        conn.close()

```

IDLE Shell 3.12.6

File Edit Shell Debug Options Window Help

Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

```

>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/client_masterA.py
('019001', datetime.date(2002, 6, 12))
('019002', datetime.date(2002, 6, 25))
('019003', datetime.date(2002, 4, 3))
('019008', datetime.date(2002, 5, 24))
('046865', datetime.date(2002, 5, 20))
('046866', datetime.date(2002, 5, 20))

```

g. List the names, city and state of clients who are not in the state of 'Maharashtra'.

```

mysql> SELECT name, city, state
-> FROM client_master
-> WHERE state != 'Maharashtra';

```

name	city	state
Client Name	Client City	Client State
Ashwini Joshi	Bangalore	Karnataka
Deepak Sharma	Mangalore	Karnataka

3 rows in set (0.00 sec)

```

import mysql.connector

try:
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="cdac123",
        database="testdb"
    )
    cursor = conn.cursor()

    cursor.execute("SELECT name, city, state FROM client_master WHERE state != 'Maharashtra';")
    results = cursor.fetchall()

    for row in results:
        print(row)

except mysql.connector.Error as err:
    print(f"Error: {err}")
finally:
    if 'cursor' in locals():
        cursor.close()
    if 'conn' in locals() and conn.is_connected():
        conn.close()

```

Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

```

>>> = RESTART: C:/Users/HP/OneDrive/Desktop/client_masterA.py
('Client Name', 'Client City', 'Client State')
('Ashwini Joshi', 'Bangalore', 'Karnataka')
('Deepak Sharma', 'Mangalore', 'Karnataka')

```

4. Exercises on Using Having, Group By and Joins in Python IDLE:

a. Printing the description and total quantity sold for each product.

```

mysql> SELECT p.DESCRPTION, SUM(sd.QTYORDERED) AS total_quantity_sold
-> FROM product_master p
-> JOIN sales_order_details sd ON p.PRODUCTNO = sd.PRODUCTNO
-> GROUP BY p.DESCRPTION;

```

DESCRIPTION	total_quantity_sold
T-Shirts	34
Denim Shirts	3
Pull Overs	5
Trousers	3
Shirts	6
Cotton Jeans	1
Lycra Tops	6

7 rows in set (0.00 sec)

```

import mysql.connector

try:

    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="cdac123",
        database="testdb"
    )
    cursor = conn.cursor()

    product_query = """
    SELECT p.DESCRPTION, SUM(sd.QTYORDERED) AS total_quantity_sold
    FROM product_master p
    JOIN sales_order_details sd ON p.PRODUCTNO = sd.PRODUCTNO
    GROUP BY p.DESCRPTION;
    """

    cursor.execute(product_query)
    product_results = cursor.fetchall()

    print("Product Description and Total Quantity Sold:")
    for row in product_results:
        print(f"Description: {row[0]}, Total Quantity Sold: {row[1]}")

except mysql.connector.Error as err:
    print(f"Error: {err}")
finally:
    if 'cursor' in locals():
        cursor.close()
    if 'conn' in locals() and conn.is_connected():
        conn.close()

```

for more information.

```

= RESTART: C:/Users/HP/OneDrive/Desktop/query.py
Product Description and Total Quantity Sold:
Description: T-Shirts, Total Quantity Sold: 34
Description: Denim Shirts, Total Quantity Sold: 3
Description: Pull Overs, Total Quantity Sold: 5
Description: Trousers, Total Quantity Sold: 3
Description: Shirts, Total Quantity Sold: 6
Description: Cotton Jeans, Total Quantity Sold: 1
Description: Lycra Tops, Total Quantity Sold: 6

```

b. Calculating the average quantity sold for each client that has a maximum order value of 15000.00.

```

mysql> SELECT c.CLIENTNO, AVG(sd.QTYORDERED) AS average_quantity_sold
-> FROM client_master c
-> JOIN sales_order s ON c.CLIENTNO = s.CLIENTNO
-> JOIN sales_order_details sd ON s.ORDERNO = sd.ORDERNO
-> GROUP BY c.CLIENTNO
-> HAVING AVG(sd.QTYORDERED * sd.PRODUCTRATE) <= 15000.00;

```

CLIENTNO	average_quantity_sold
C00001	2.2000
C00002	10.0000
C00004	3.6667
C00005	7.5000

4 rows in set (0.01 sec)

```

import mysql.connector

try:
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="cdac123",
        database="testdb"
    )
    cursor = conn.cursor()

    average_quantity_query = """
    SELECT c.CLIENTNO, AVG(sd.QTYORDERED) AS average_quantity_sold
    FROM client_master c
    JOIN sales_order s ON c.CLIENTNO = s.CLIENTNO
    JOIN sales_order_details sd ON s.ORDERNO = sd.ORDERNO
    GROUP BY c.CLIENTNO
    HAVING AVG(sd.QTYORDERED * sd.PRODUCTRATE) <= 15000.00;
    """

    cursor.execute(average_quantity_query)
    average_quantity_results = cursor.fetchall()

    print("\nClient Number and Average Quantity Sold:")
    for row in average_quantity_results:
        print(f"Client Number: {row[0]}, Average Quantity Sold: {row[1]}")

except mysql.connector.Error as err:
    print(f"Error: {err}")
finally:
    if 'cursor' in locals():
        cursor.close()
    if 'conn' in locals() and conn.is_connected():
        conn.close()

```

= RESTART: C:/Users/HP/OneDrive/Desktop/query.py

```

Client Number and Average Quantity Sold:
Client Number: C00001, Average Quantity Sold: 2.2000
Client Number: C00002, Average Quantity Sold: 10.0000
Client Number: C00004, Average Quantity Sold: 3.6667
Client Number: C00005, Average Quantity Sold: 7.5000

```